2020 线性代数期中(回忆版).

1. Multiple Choice. Only one choice is correct.

(1) Let A be $m\times n$ matrix and b be a column vector in $\mathbb{R}^m$. Which of the following statements is correct? ( )

(A) If $Ax=b$ has infinite many solutions, then $Ax=0$ has a nonzero solution.

(B) If the system $Ax=0$ has only zero solution, then $Ax=b$ has one and only one solution.

(C) If the rank of A is n, then the system $Ax=b$ must have a solution.

(D) If A is a square matrix (i.e, $m=n$), then the system $Ax=b$ is consistent if and only if A is invertible.

(2) Suppose A is an $m\times n$ matrix, B is an $n\times m$ matrix, and I is the $m\times$~~n~~ m identity matrix. If $AB=I$, then ( ).

(A) the column vectors of A are linearly independent, and the row vectors of B are linearly independent.

(B) the column vectors of A are linearly independent, and the column vectors of B are linearly independent.

(C) the row vectors of A are linearly independent, and the column vectors of B are linearly independent.

(D) the row vectors of A are linearly independent, and the row vectors of B are linearly independent.

(3) Let A be a 3×3 matrix, and let B be (the) the matrix formed by adding the second column of A to its first column. Suppose that after exchanging the second and third rows of B, the resulting matrix is the 3×3 identity matrix. Let $P_1=\begin{bmatrix}1&0&0\\1&1&0\\0&0&1\end{bmatrix}$, $P_2=\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$. Then $A=$ ( )

(A) P_1P_2 (B) $P_1^{-1}P_2$ (C) P_2P_1 (D) $P_2P_1^{-1}$.

(4) Let $A=\begin{bmatrix}-1&2&3\\-3&6&8\\2&-4&t\end{bmatrix}$, where $t\in R$. Suppose rank $(A)=2$. Then ( ).

(A) $t=-6$ (B) $t=6$ (C) $t\neq0$ (D) t can be any real number.

(5) Which of the following statement is ~~c~~ incorrect? ( )

(A) For any matrix A, rank $(A)=\dim C(A)$.

(B) If $v_1,\cdots,v_m$ are pairwise orthogonal nonzero vectors, then the vectors $v_1,\cdots,v_m$ are linear independent.

(C) If A is an upper triangular $n\times n$ matrix such that $A^2=0$, then $A=0$.

(D) Let A,B be $n\times n$ matrices such that AB is invertible. Then both A and B are ~~inter~~-invertible.

2. Fill in the blanks.

(1) Let A, B be invertible $n \times n$ matrices. Then the inverse of the block matrix $\begin{bmatrix} 0 & A \\ B & 0 \end{bmatrix}$ is _____.

(2) Suppose A is a 3×4 matrix and $\dim N(A) = 2$. Then $\dim N(A^T) =$ _____.

(3) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}$. Then $A^{-1} =$ _____.

(4) Let u, v be vectors in R^n such that $\|u\| = 3$, $\|v\| = 4$ and $u^T v = -3$.

Then $\|2u + 3v\| =$ _____.

(5) Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}$. Then the least squares solution to $Ax = b$ is $\hat{x} =$ _____.

3. Find the LU factorization of the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$.

4. Let $A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$. Please give a basis ofor each of the four fundamental subspaces $C(A)$, $N(A)$, $C(A^T)$ and $N(A^T)$.

5. Let $E = \{u_1, u_2, u_3\}$ and $F = \{v_1, v_2\}$, where

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, u_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \text{ and } v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Define the linear transformation $T: R^3 \to R^2$ by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 2x_2 \\ -x_1 \end{bmatrix}$$

Find the matrix A representing T with respect to the ordered bases E and F.

6. Let A, B be $n \times n$ matrices. Suppose A and B are both symmetric. Is AB necessarily symmetric? If yes, please give a proof. Otherwise please give a counterexample.

7. The following two questions are independent:

(a) Let A be the 2×2 matrix such that the linear transformation $R^2 \to R^2$, $v \mapsto Av$ rotates every vector in R^2 through $60°$ counter-clockwise (about the origin). Find A and A^{2020}.

(b) Three planes Π_1, Π_2, Π_3 in the space R^3 are given by the equations

$$\Pi_1: x + y + z = 0,$$
$$\Pi_2: 2x - y + 4z = 0,$$
$$\Pi_3: -x + 2y - z = 0.$$

Determine a matrix representative (in the standard basis of R^3) of a linear transformation taking the xy plane to Π_1, the yz plane to Π_2 and the zx plane to Π_3.

8. Let A be a 3×3 matrix such that $\text{rank}(A)=2$ and $A^3=0$.

(a) Prove that $\text{rank}(A^2)=1$.

(b) Let $\alpha_1 \in R^3$ be a nonzero vector such that $A\alpha_1=0$. Prove that there exist vectors α_2, α_3 such that $A\alpha_2=\alpha_1$, $A^2\alpha_3=\alpha_1$.

(c) For any vectors α_2, α_3 described as above, show that $\alpha_1, \alpha_2, \alpha_3$ are linearly independent.

(In this problem, you are allowed to assume the statements of some questions to answer subsequent questions.)